Boris Kafidov

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Education

University of Toronto — Toronto, ON
Honours B.Sc. in Computer Science & Mathematics, Focus in Artificial Intelligence — *Sep. 2024 – Jun. 2028*

Experience

Amazon — Toronto, ON
Jr. Software Development Engineer (Internship) — *Jun. 2026 – Present*

- Root-caused and eliminated product data discrepancies across three regions by supplying missing marketplace context to product API requests from a customer-facing TypeScript frontend
- Authored the technical design for a 12-marketplace TypeScript product-page frontend migration and led a five-engineer review comparing API- and search-backed architectures for scalability and data ownership
- Developed TypeScript AWS CDK definitions for 18 CloudWatch alarms across three regions and desktop, mobile, and tablet surfaces, designed to replace six manual monitors with sustained escalation and incident routing
- Shipped TypeScript, semantic HTML, and CSS fixes for three accessibility defects spanning keyboard activation, button semantics, and high-contrast visibility; escalated the shared-framework root cause to its owning team

University of Toronto, Wheeler Lab — Toronto, ON
Software Engineer (Work Study) — *Oct. 2025 – Mar. 2026*

- Built simulated ASI stage and camera backends for a Python/PySide6 Linux microscopy GUI, enabling hardware-independent development; fixed simulated movement signaling that triggered false 10-second tiling waits
- Instrumented camera, stage, preset, and per-channel acquisition paths with buffered JSONL profiling, recording run IDs, timestamps, wait modes, timeouts, and per-tile duration to isolate bottlenecks during microscope runs
- Reduced repeated device waits during multi-channel acquisition by issuing cube, excitation, and intensity updates in dependency order, then waiting once on the group with sequential fallback on timeout

Research

A 13-vertex counterexample to e-log-concavity for chromatic quasisymmetric functions — *Jul. 2026*
Sole-authored arXiv preprint. Exact, computer-assisted disproof of Abreu–Nigro Conjecture 5.3

Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma — *Jul. 2026*
Sole-authored arXiv preprint. Proves the dimension-16 case via matrix scaling and permanent bounds

Projects

CreditCombo | *Live App* | *JavaScript, Web Workers, Cloudflare Workers* — *Feb. 2026 – Mar. 2026*

- Built and launched a browser-based optimizer that selects Canadian credit-card portfolios and category-level usage strategies by modeling net annual value across 151 cards, 51 rewards programs, and 22 spend subcategories
- Implemented deterministic, cap-aware search that annualizes spend, prices rewards under estimated and guaranteed valuations, reroutes shared-cap overflow, and honors user-locked cards and annual-fee caps
- Built a responsive browser-only runtime that executes combinatorial searches in a Web Worker, discards stale results via request IDs, and uses a bounded cache; deployed through Cloudflare with route rewrites and CSP

APA Site Choice Transformer | *Python, PyTorch, Hugging Face, GCP, AWS, Git LFS* — *Sep. 2025*

- Co-led a 5-person undergraduate team to 2nd place among 23 projects and 116 contestants, earning a $500 award
- Optimized DNABERT-2-117M training on 800k labeled DNA/RNA windows using A100 GPUs on GCP with mixed precision and gradient accumulation; offloaded evaluation logits to CPU to prevent OOM failures
- Prevented train/eval leakage by holding out entire chromosomes and standardizing genomic coordinates across PolyASite 2.0, PolyA_DB, and GENCODE with genome_kit before extracting sequence windows
- Achieved 0.86 accuracy, 0.93 AUROC, 0.94 AUPRC, and 0.86 F1 on held-out chromosomes using sequence alone

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C
Frameworks/Libraries: PyTorch, Hugging Face Transformers, Node.js, scikit-learn, NumPy, Pandas, PySide6, Qt
Tools/Platforms: Git, Linux, Bash, AWS CDK, CloudWatch, GCP, Cloudflare Workers, Git LFS
Concepts: Data Structures & Algorithms, REST APIs, Unit Testing, Accessibility, Infrastructure as Code